Mason Alexander Headley

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Engineering Profile:

Electrical Engineering student pursuing a Computer Science minor and an Aerospace Engineering certificate, with Private Pilot License and three consecutive summers of technical industry experience. Combines hands-on experience in electrical systems, engineering design and manufacturing, CAD, programming, and avionics with demonstrated leadership and project management in aerospace projects and flight operations.

Education, Certifications, and Licenses:

- **University of Georgia** - B.S. Electrical Engineering | Minor Computer Science | Aerospace Engineering Certificate | Zell Miller Scholarship | *January 2024 - Present* | 3.3 GPA | Graduation Spring 2028
  Relevant coursework: Electronics I & II, Sensors and Transducers, Linear Systems, Electromagnetics, Power Systems Analysis I & II, Aerospace Systems Engineering, Feedback Control Systems | Complete or Completed by Summer 2027
- **FAA Private Pilot Certificate -**| ASEL | 80+ hours
- **FAA Remote Pilot Certificate -**| in progress
- **Eagle Scout -** Scouting America | Quartermaster

Professional Experience:

- **Parallax Defense - Hardware Team (Fall 2026 - present)**
  Testing and evaluating advanced sensor payloads (LiDAR, Radar, EO/IR); Designing, building, and flying target drone / rig for product tracking development and demonstration
- **Trackmobile - Engineering Intern (Summers 2025 and 2026)**
  Created and updated drawings and installation instructions for purchasing options; Utilized Zuken for electrical harness design and Solidworks integration; Applied DDM Office file management and revision control
- **Shenandoah Electrical - Electrician's Assistant (Summer 2024)**
  Installed and wired high and low voltage outlets and power hookups; Interpreted electrical and lighting plans and schematics to ensure accurate installations
- **Delta Air Lines - Cloud Security Engineer (Summer 2023)**
  Developed a CFN Guard rule to encrypt traffic to origin; Tested commands in AWS CloudShell CLI and documented technical findings into sharepoint

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Aerospace Projects & Leadership:

- **AIAA Design Build Fly** - Avionics Sub-Team Lead | collaborated on avionics design and testing
- **AIAA RC Planes Team Lead** - 3 years | Grew active membership from 5 to 30+
- **Independent RC projects** - building and upgrading RC cars, boats, FPV quadcopters, and planes

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Skills:

- **Programming**: Java, Python, CSS, HTML, C, C++, Git, GitHub, G-code, MATLAB
- **CAD & Design**: Solidworks 25, KiCad, LTspice, Multisim, Onshape, Zuken, Visual Studio Code, Arduino IDE, 3D printing, Microsoft Office, and Google Workplace
- **Electrical Engineering**: Circuit design, PCB design, Electrical Harness design, Soldering, ESP32, Raspberry pi, Arduino, Oscilloscope, Digital Multimeter, Breadboard Prototyping, Power Systems
- **Cloud & Security**: AWS (CloudShell CLI, CFN Guard), Networking Basics, 2FA setup